

A Complete Counting Characterization of Schatten Rhaly Matrices for Every Positive Exponent

Full solution packet for arXiv:2503.21930

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Status: full solution likely valid. For every $q > 0$, membership of a Rhaly matrix R_α in $\mathcal{S}^q(\ell^2)$ is characterized by the q -moment of the length of the source paper's greedy (ε, L) -sequence. Explicit two-sided quasinorm estimates are proved. In particular, this answers the source paper's missing-characterization question for the entire range $0 < q \leq 1$.

1 Source question

Bellavita, Dellepiane, and Stylogiannis study the Rhaly matrix

$$(R_\alpha x)(k) = \alpha_k \sum_{j=0}^k x(j), \quad k \in \mathbb{N}, \quad (1)$$

for $\alpha = (\alpha_k) \in \ell^2$ [1]. Their Theorem 1.3 gives a dyadic coefficient characterization of $R_\alpha \in \mathcal{S}^q$ when $1 < q < \infty$. Theorem 5.5 gives the corresponding dyadic condition only as a necessary condition for arbitrary $q > 0$. In the conclusion, on source PDF page 36, they state:

“Theorem 5.5 only provides a necessary condition for the q -Schatten class membership, for $0 < q \leq 1$. A complete characterization is missing.”

7. CONCLUSION

The (ε, L) -sequence technique may be useful for studying approximate values of other significant operators. First of all, one shall consider the factorable matrices, already defined in (2.3), which depend on two sequences α_n and β_n . The arguments presented in this article may adapt also to these operators. In this situation, the conditions on f_α will be in terms of weighted Besov spaces.

There are still several open questions related to the Rhaly matrices. First of all, Theorem 5.5 only provides a necessary condition for the q -Schatten class membership, for $0 < q \leq 1$. A complete characterization is missing.

Figure 1: The endpoint open question in arXiv:2503.21930v3, source PDF page 36.

2 The coefficient-side counting invariant

For a natural interval I and x supported in I , put

$$S_x(k) = \sum_{j=0}^k x(j), \quad \mu_\alpha(I) = \sum_{k \in I} |\alpha_k|^2.$$

When $\mu_\alpha(I) > 0$, let

$$(S_x)_I = \frac{1}{\mu_\alpha(I)} \sum_{k \in I} S_x(k) |\alpha_k|^2$$

and define

$$L_\alpha(I) = \sqrt{2} \sup_{\substack{\text{supp } x \subseteq I \\ \|x\|_2 \leq 1}} \left(\sum_{k \in I} |S_x(k) - (S_x)_I|^2 |\alpha_k|^2 \right)^{1/2}. \quad (2)$$

Set $L_\alpha(I) = 0$ if $\mu_\alpha(I) = 0$, and define $L_\alpha([a, \infty))$ by increasing finite truncations. Formula (2) is the equivalent form of the source paper's interval invariant given by its Lemma 3.6.

For $\varepsilon > 0$, start with $c_0 = 0$ and recursively set

$$c_{r+1} = \inf\{t \in \mathbb{N} : t > c_r, L_\alpha([c_r, t-1]) > \varepsilon\}. \quad (3)$$

If $c_N < \infty$ and $c_{N+1} = \infty$, the sequence has length N ; if it never terminates, its length is ∞ . Denote this length by $N_\alpha(\varepsilon)$. This is entirely determined by the coefficients α . It is an extended integer-valued Borel function of ε : for fixed m , the event $N_\alpha(\varepsilon) \geq m$ is a countable union of conditions involving finitely many comparisons $L_\alpha([a, b]) > \varepsilon$.

For $q > 0$, define the endpoint counting functional

$$\Phi_q(\alpha) = q \int_0^\infty \varepsilon^{q-1} N_\alpha(\varepsilon) d\varepsilon \in [0, \infty]. \quad (4)$$

Also set

$$\sigma_{-1} = |\alpha_0|, \quad \sigma_k = \left(\sum_{j=2^k}^{2^{k+1}-1} (j+1) |\alpha_j|^2 \right)^{1/2}, \quad M_\alpha = \sup_{k \geq -1} \sigma_k. \quad (5)$$

The source paper's Theorem 2.4 says that $M_\alpha < \infty$ exactly when R_α is bounded.

3 Proof intuition

The greedy length already measures the whole singular-value distribution, not merely compactness. The source paper proves two estimates. If the ε -greedy sequence has length N , then the rank of a natural piecewise-constant approximation is at most $2N + 1$, so $a_{2N+2}(R_\alpha) \leq \varepsilon/\sqrt{2}$. Conversely, N disjoint greedy intervals force $a_N(R_\alpha) \geq \varepsilon/\sqrt{2}$.

Thus the number of approximation values above a threshold is trapped between the same greedy length, with only a factor two in the count and fixed rescalings of the threshold. Integrating this distributional sandwich via the layer-cake identity gives the complete $\mathcal{S}^q$ criterion for every $q > 0$. The endpoint obstruction in the source proof disappears because we do not dominate the greedy count by an additional harmonic sum over dyadic levels; that domination is precisely where the restriction $q > 1$ entered.

4 Distributional sandwich

Let $a_m = a_m(R_\alpha)$ be the approximation numbers, indexed so that $a_1 = \|R_\alpha\|$, and put

$$n_\alpha(t) = \#\{m \geq 1 : a_m > t\}.$$

Both sides are allowed to be infinite.

Lemma 1 (source estimates in counting form). *For every $\varepsilon > 0$,*

$$N_\alpha(\varepsilon) \leq n_\alpha\left(\frac{\varepsilon}{2\sqrt{2}}\right), \quad n_\alpha\left(\frac{\varepsilon}{\sqrt{2}}\right) \leq 2N_\alpha(\varepsilon) + 1. \quad (6)$$

Proof. Suppose first that $N_\alpha(\varepsilon) = N < \infty$. If $N \geq 1$, source Lemma 4.3 gives

$$a_N \geq \frac{\varepsilon}{\sqrt{2}} > \frac{\varepsilon}{2\sqrt{2}}.$$

Hence at least N approximation numbers exceed $\varepsilon/(2\sqrt{2})$. The first assertion is trivial when $N = 0$. If the greedy sequence is infinite, source Lemma 4.3 applies with every finite N , so the first counting function is infinite as well.

For the other direction, source Lemma 4.2 gives

$$a_{2N+2} \leq \frac{\varepsilon}{\sqrt{2}}$$

when the length is $N < \infty$. Therefore at most $2N + 1$ approximation numbers are strictly larger than $\varepsilon/\sqrt{2}$. If the length is infinite, the claimed inequality is automatic. $\square$

5 Full characterization

Theorem 2 (all positive Schatten exponents). *Let $q > 0$ and suppose R_α is bounded on ℓ^2 . With the convention that all quantities may equal $+\infty$,*

$$\sum_{m=1}^{\infty} a_m(R_\alpha)^q \leq \|R_\alpha\|^q + 2^{1-q/2} \Phi_q(\alpha), \quad (A)$$

$$\|R_\alpha\|^q + \Phi_q(\alpha) \leq (1 + (2\sqrt{2})^q) \sum_{m=1}^{\infty} a_m(R_\alpha)^q. \quad (B)$$

Consequently,

$$R_\alpha \in \mathcal{S}^q(\ell^2) \iff \Phi_q(\alpha) < \infty. \quad (7)$$

Proof. For every nonincreasing nonnegative sequence (a_m) and every $q > 0$, Tonelli's theorem gives the extended layer-cake identity

$$\sum_{m=1}^{\infty} a_m^q = q \int_0^\infty t^{q-1} \#\{m : a_m > t\} dt. \quad (8)$$

Integrating the first inequality in (6), substituting $t = \varepsilon/(2\sqrt{2})$, and using (8), we obtain

$$\Phi_q(\alpha) \leq (2\sqrt{2})^q \sum_{m=1}^{\infty} a_m^q.$$

Since $\|R_\alpha\| = a_1$, this proves (B).

For the reverse estimate, put $\varepsilon = \sqrt{2}t$ in the second inequality of (6). Since $n_\alpha(t) = 0$ for $t \geq \|R_\alpha\|$, another application of (8) gives

$$\begin{aligned} \sum_{m=1}^{\infty} a_m^q &\leq 2q \int_0^{\|R_\alpha\|} t^{q-1} N_\alpha(\sqrt{2}t) dt + q \int_0^{\|R_\alpha\|} t^{q-1} dt \\ &\leq 2^{1-q/2} \Phi_q(\alpha) + \|R_\alpha\|^q. \end{aligned}$$

This is (A), and the equivalence follows. $\square$

Combining the theorem with the source boundedness criterion yields a statement containing only coefficient-side quantities.

Corollary 3 (complete coefficient characterization). *Let $\alpha \in \ell^2$ and $q > 0$. Then*

$$R_\alpha \in \mathcal{S}^q(\ell^2) \iff M_\alpha < \infty \text{ and } \Phi_q(\alpha) < \infty,$$

where M_α and $\Phi_q(\alpha)$ are given explicitly by (2)–(4) and (5). In particular, this is a complete characterization for every $0 < q \leq 1$.

Proof. Theorem 2.4 of the source gives $M_\alpha < \infty$ if and only if R_α is bounded. On that boundedness domain, Theorem 2 applies. Every $\mathcal{S}^q$ operator is bounded, so the two conditions are also necessary. $\square$

6 Verification, scope, and novelty

The proof has no computational dependency. The important endpoint details are the strict threshold in $n_\alpha(t)$, the harmless factor two in the first half of (6), and the separate operator-norm term in (A). That last term cannot simply be discarded: if only α_0 is nonzero, all greedy lengths vanish while R_α is a nonzero rank-one operator.

The result gives an exact greedy-counting characterization, not a reduction to the simpler dyadic Besov expression used by the source for $q > 1$. Thus it answers the stated request for a complete characterization, while leaving open whether a comparably simple dyadic coefficient formula exists at the quasi-Banach endpoints.

A bounded novelty search on 22 July 2026 covered the run’s four lightweight indexes; the exact arXiv id and title; searches for Rhaly/terraced matrices with Schatten endpoints, approximation numbers, and (ε, L) sequences; the current source version; and the June 2026 follow-up arXiv:2606.15875. That follow-up extends the method to factorable matrices but still states its Schatten characterizations only for $1 < p < \infty$. No later source stating the counting criterion above or resolving the $0 < q \leq 1$ question was found. Novelty confidence is moderate.

Human-review recommendation. Audit the interpretation of the source’s length convention at $N = 0$ and $N = \infty$, then verify that Lemmas 4.2 and 4.3 apply with precisely the approximation-number indexing used in (6). Once that interface is confirmed, the remaining proof is Tonelli integration.

References

- [1] C. Bellavita, E. Dellepiane, and G. Stylogiannis, *Boundedness, compactness and Schatten class for Rhaly matrices*, arXiv:2503.21930v3 (2025), especially Definition 4.1, Lemmas 4.2–4.3, Theorem 2.4, Theorem 5.5, and Section 7.